Ultrasonic-PID-Motor-Control

Arduino Real-Time Implementation using Simulink

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Simulink Model: Attached in the Repository

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1. Introduction

This project implements a closed-loop motor speed control system that integrates an ultrasonic sensor, encoder gear motor, PID controller, and Arduino Uno to achieve precise and adaptive motion control. The ultrasonic sensor continuously measures the distance to an obstacle and serves as the variable reference input, allowing the system to automatically adjust motor speed based on environmental conditions. Encoder feedback provides accurate, real-time information about motor rotation, enabling the PID controller to correct speed deviations and maintain stable performance.

Simulink is used to model the control logic, tune the PID parameters, visualize system behavior, and automatically generate code for deployment onto the Arduino. This end-to-end workflow demonstrates key engineering competencies, including dynamic system modeling, sensor integration, embedded implementation, and real-time data collection. The project showcases how low-cost hardware combined with Simulink-based control design can produce a reliable, efficient, and scalable motor control solution suitable for robotics and automation applications.

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2. System Overview

2.1. Circuit Connections

The system uses an HC-SR04 ultrasonic sensor for real-time distance measurement, an encoder gear motor for precise speed and position feedback, an L298N dual H-bridge motor driver for bidirectional motor actuation, and an Arduino Uno to execute the PID control algorithm and interface with all components. The encoder provides pulse-based speed feedback that enables accurate closed-loop correction, while the ultrasonic sensor determines the desired motor response based on obstacle distance. A 9V external power supply is used to drive the motor through the L298N, ensuring sufficient current without overloading the Arduino. Together, these components form a reliable and modular hardware platform capable of real-time sensing, control, and actuation.

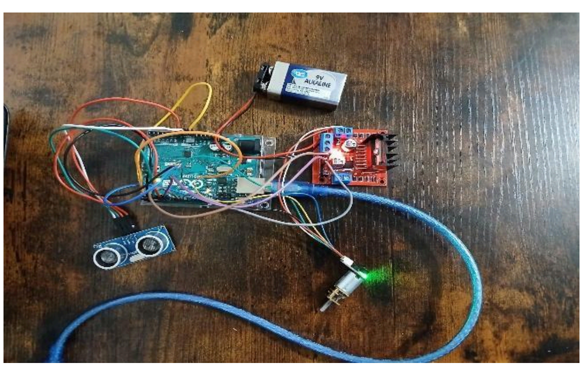

Figure Circuit Connections

2.2. Simulink Model

The Simulink model implements a closed-loop control system where the ultrasonic sensor input is read through Arduino and compared against a threshold to determine the desired response. The measured distance passes through a switch and reference logic to calculate the system error, which is fed into a PID controller. The PID block generates a corrective output that is scaled and limited through a gain and saturation block before being converted into a PWM signal sent to Arduino Pin 9. This PWM signal controls the motor speed, while the scopes display real-time error and control input. Overall, the model continuously adjusts the motor speed based on distance, ensuring stable and accurate PID-based control.

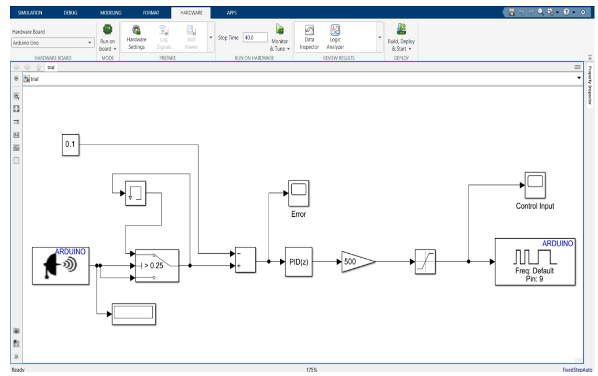

Figure 2 Simulink Model

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3. Operation

3.1. Before Running

Before executing the Simulink model, the system remains in an open-loop state. No distance values appear on the display because the ultrasonic sensor has not yet begun real-time measurement. The Error and Control Input scopes show no waveforms, indicating that the PID controller is not active. During this phase, the motor runs at a fixed, uncontrolled speed, driven only by its default power input. This confirms that the hardware is correctly connected and powered prior to enabling the closed-loop control logic.

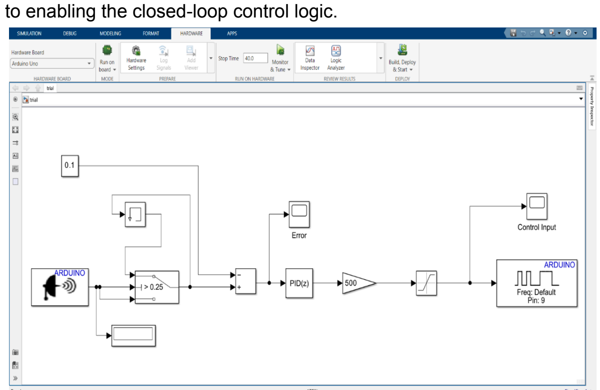


Figure 3 Before Running

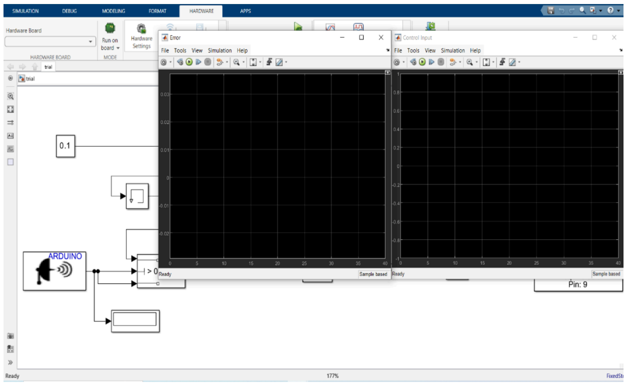
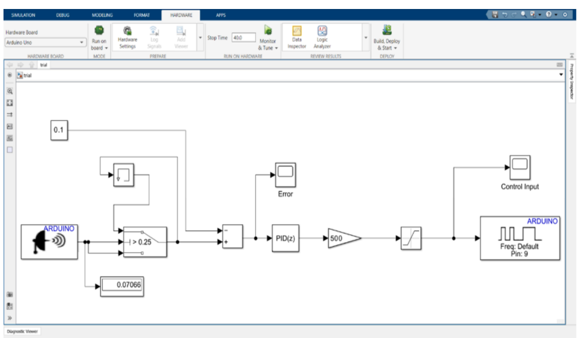


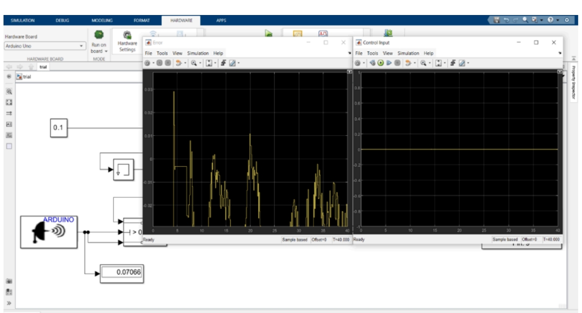
Figure 4 Before Running Output

3.2. After Running

Once the Simulink model is deployed and running, the system enters closed-loop operation. Distance readings update continuously from the ultrasonic sensor and are fed into the control algorithm. The Error scope becomes active as the system computes deviation from the setpoint in real time. When the measured distance is greater than 5 cm, the PID controller increases the PWM output, causing the motor to speed up; when the distance is less than 5 cm, the controller reduces the PWM signal, slowing the motor. Around the target distance of 5 cm, the motor stabilizes with minimal oscillation due to effective PID regulation. This confirms successful real-time closed-loop speed control.



(Figure 5 After Running)



(Figure 6 After Running Output)

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4. Results

The PID controller effectively minimizes speed error and maintains stable motor operation across varying distances. The motor responds smoothly and predictably to changes in the ultrasonic sensor input, demonstrating strong dynamic performance. The integration of Simulink with the Arduino enables reliable real-time execution, and data captured from the scopes and display blocks confirms accurate distance sensing, consistent feedback behavior, and precise closed-loop speed regulation.

Video Link

Attached in the Repository.

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5. Advantages, Limitations & Applications

5.1. Advantages

1. Provides accurate and stable speed control through encoder feedback.

2. Adapts motor speed real time on environmental distance changes.

3. Uses low-cost, modular hardware, easy to upgrade and scale.

4. Ensures smooth and responsive motor performance through effective PID regulation.

5.2. Limitations

1. PID tuning may require multiple iterations to achieve optimal performance.

2. Ultrasonic sensor accuracy can be affected by surface angle, noise, and environmental conditions.

3. The L298N motor driver has limited current capacity, restricting use with larger motors.

4. System performance depends on consistent encoder feedback; any misalignment or signal noise affects accuracy.

5.3. Applications

1. Robotics systems requiring obstacle-aware speed control.

2. Conveyor and material-handling automation.

3. Autonomous and semi-autonomous navigation platforms.

4. Smart home motorized devices (e.g., automated doors, fans, curtains).

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6. Conclusion

This project successfully demonstrates a compact and cost-effective closed-loop motor speed control system using ultrasonic sensing, encoder feedback, and PID regulation. The real-time Simulink-Arduino integration enables precise, adaptive, and stable motor behavior in response to changing distances. Overall, the system validates the effectiveness of combining low-cost hardware with model-based design tools and provides a practical foundation for a wide range of automation and robotics applications.